Galactica : A Large Language Model for Science

Focus on past techniques and research

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The background behind the development of the Galactica


AS WE MAY THINK

A TOP U. S. SCIENTIST FORESEES A POSSIBLE FUTURE WORLI IN WHICH MAN-MADE MACHINES WILL START TO THINI

by VANNEVAR BUSH

「LIFE」1945年9月10日号に特集された「AS WE MAY THINK」のページ

「LIFE」1945年9月10日号より引用

「AS WE MAY THINK」 From the Atlantic Monthly, July 1945

- Computers are a solution to manage the growing mountain of information.
- Computers would take care of routine tasks such as storage and retrieval, "preparing the way for insights and decisions in scientific thinking"

information overload remains an overwhelming problem!!

it is impossible for a single humans to read and process vast amounts of data in a usable form. to organize knowledge directly, Restructuring the data is challenging.

The background behind the development of the Galactica

- Search engines are the current interface for accessing scientific knowledge.
- No matter how powerful search engines are, they still rely on human manual work, which slows down information utilization and overwhelms researchers.
- Unlike search engines, language models can store, combine and reason scientific knowledge.
- Galactica is trained on a large and curated corpus of humanity’s scientific knowledge. It could synthesize knowledge by generating secondary content automatically.

Past research and techniques

인용된 논문 소개

pher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. *CoRR*, abs/2005.14165, 2020. URL https://arxiv.org/abs/2005.14165.

GPT - 3

Geoffrey Irving. Scaling language models: Methods, analysis & insights from training gopher. *CoRR*, abs/2112.11446, 2021. URL https://arxiv.org/abs/2112.11446.

Gopher

Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022. URL https://arxiv.org/abs/2203.15556.

Chinchilla

Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. Opt: Open pre-trained transformer language models, 2022. URL https://arxiv.org/abs/2205.01068.

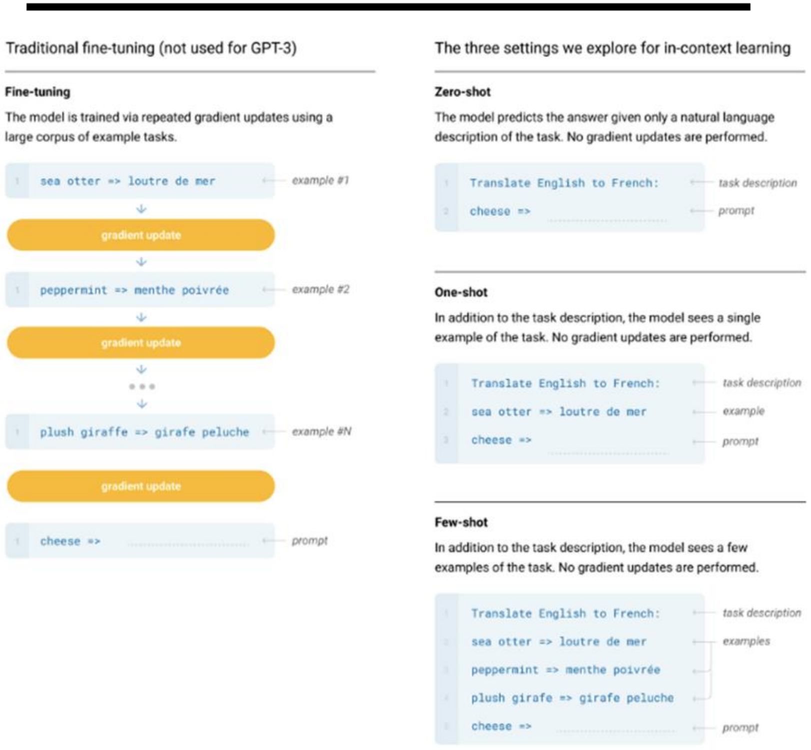
Opt

Jeff Dean, Slav Petrov, and Noah Fiedel. Palm: Scaling language modeling with pathways, 2022. URL https://arxiv.org/abs/2204.02311.

PaLM

Past research and techniques

1. GPT–3 (2020, OpenAI)



datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

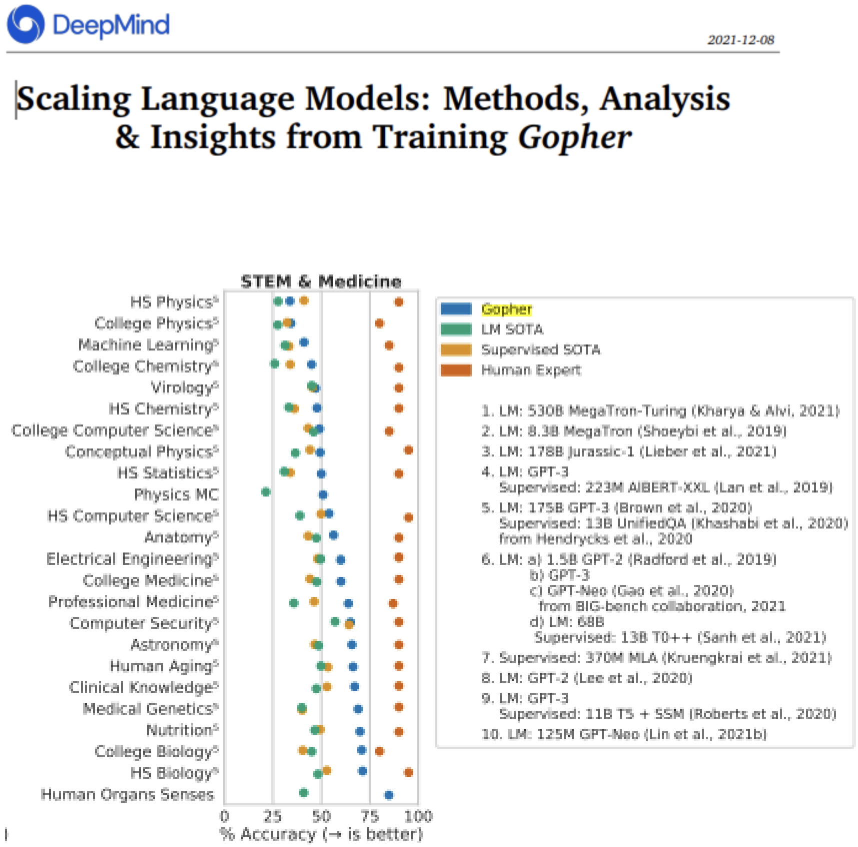
- GPT-3 is a large language model designed to perform various natural language processing tasks using few-shot, one-shot, or even zero-shot learning.

- It consists of 175 billion parameters, making it one of the largest models of its time. GPT-3 demonstrated the ability to generalize across tasks without task-specific fine-tuning by leveraging prompts, allowing it to perform tasks like translation, question answering, and text generation with minimal examples.

- Its introduction marked a significant step forward in demonstrating the power of large-scale models for few-shot learning.

Past research and techniques

2. Gopher (2021, DeepMind)



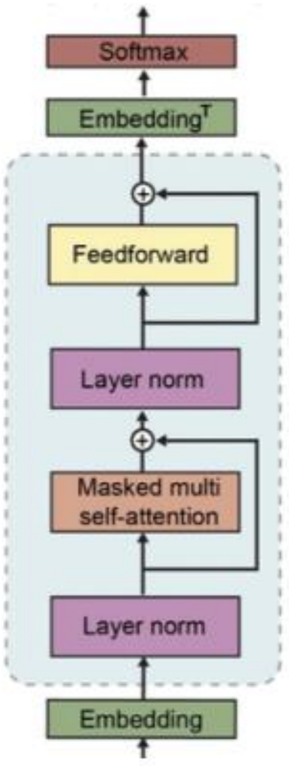
Gopher is a 280 billion parameter large-scale language model designed for state-of-the-art performance.

Trained on a high-quality dataset called MassiveText, with a comprehensive training protocol covering architecture design, optimization, and infrastructure.

While Gopher demonstrates significant performance improvements and potential, the paper emphasizes the need to address challenges such as toxicity and bias to ensure ethical and safe usage.

Past research and techniques

2. Gopher (2021, DeepMind)



GPT-2 decoder block

$$\bar{a}_i = \frac{a_i}{\text{RMS}(\mathbf{a})} g_i,$$

$$\text{where} \;\; \text{RMS}(\mathbf{a}) = \sqrt{\frac{1}{n}\sum_{i=1}^{n} a_i^2}.$$

Gopher made two modifications to the existing autoregressive Transformer architecture:

1. Replaced LayerNorm with RMSNorm.

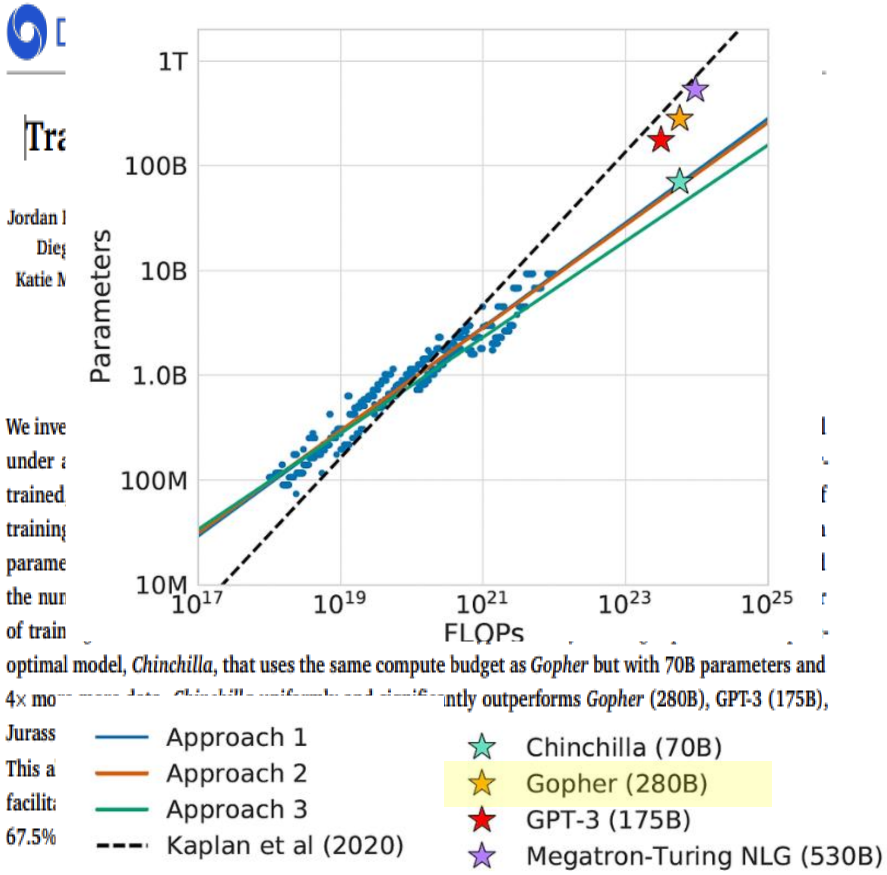
   Normalization using the root mean square (RMS):
   - Requires only the calculation of the root mean square, making it less computationally expensive.
   - Enables normalization with lower computational cost and simpler calculations.

   LN (LayerNorm): Requires the calculation of both mean and variance, resulting in higher computational cost.

2. Switched from absolute positional encoding to relative positional encoding, allowing it to evaluate longer sequences than those seen during training.

Past research and techniques

3. Chinchilla (2022, DeepMind)



Chinchilla explores the optimal **scaling law** from three perspectives.

Given a computing budget C, expressed as a function $FLOPs(N,D)$ the ultimate goal is to minimize L.

$$N_{opt}(C), D_{opt}(C) = \underset{N,D \text{ s.t. } \mathrm{FLOPs}(N,D)=C}{\operatorname{argmin}} L(N, D).$$

$L(N,D)$:The final pre-training loss
N : the number of model parameters
D : the number of training tokens used.
$Nopt$,$Dopt$: the optimal values of N and D allocated based on the computing budget.

Past research and techniques

3. Chinchilla (2022, DeepMind)

Random	25.0%
Average human rater	34.5%
GPT-3 5-shot	43.9%
Gopher 5-shot	60.0%
***Chinchilla* 5-shot**	**67.6%**
Average human expert performance	*89.8%*
June 2022 Forecast	57.1%
June 2023 Forecast	63.4%

Based on this principle, the **Chinchilla model** was designed with **70 billion parameters** and trained on **1.4 trillion tokens**, aligning with Gopher's computational budget.

This approach optimized model accuracy and resulted in **Chinchilla outperforming larger models** such as **Gopher (280B)** and **GPT-3 (175B)** by a significant margin.

	Chinchilla	*Gopher*
All	78.3%	71.4%
Male	71.2%	68.0%
Female	79.6%	71.3%
Neutral	84.2%	75.0%

In tests related to gender bias, Chinchilla demonstrates higher accuracy compared to Gopher, indicating it is less biased.

Past research and techniques

3. Chinchilla (2022, DeepMind)

Scaling Laws The idea of "scaling laws" was put forward by Kaplan et al. (2020), who demonstrated evidence that loss scales as a power-law with model size, dataset size, and the amount of training compute. The focus was on upstream perplexity, and work by Tay et al. (2022a) showed that this does not always correlate with downstream performance. Hoffmann et al. (2022) presented new analysis taking into account the optimal amount of data, and suggested that existing language models were undertrained: "Chinchilla scaling laws". This work did not take into the account of fresh versus repeated tokens. In this work, we show that we can improve upstream and downstream performance by training on repeated tokens.

Limitations of Chinchilla (as identified in the Galactica study)

This study did not consider the distinction between novel tokens and repeated tokens.
The Galactica study demonstrates that leveraging repeated tokens during training can improve both perplexity and downstream performance of the language model.

Past research and techniques

4. OPT (2022, Meta AI)

OPT: Open Pre-trained Transformer Language Models

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Abstract

Large language models, which are often trained for hundreds of thousands of compute days, have shown remarkable capabilities for zero- and few-shot learning. Given their computational cost, these models are difficult to replicate without significant capital. For the few that are available through APIs, no access is granted to the full model weights, making them difficult to study. We present Open Pre-trained Transformers (OPT), a suite of decoder-only pre-trained transformers ranging from 125M to 175B parameters, which we aim to fully and responsibly share with interested researchers. We show that OPT-175B is comparable to GPT-3,[1] while requiring only 1/7th the carbon footprint to develop. We are also releasing our logbook detailing the infrastructure challenges we faced, along with code for experimenting with all of the released models.

progress on improving known challenges in areas such as robustness, bias, and toxicity.

In this technical report, we present Open Pre-trained Transformers (OPT), a suite of decoder-only pre-trained transformers ranging from 125M to 175B parameters, which we aim to fully and responsibly share with interested researchers. We train the OPT models to roughly match the performance and sizes of the GPT-3 class of models, while also applying the latest best practices in data collection and efficient training. Our aim in developing this suite of OPT models is to enable reproducible and responsible research at scale, and to bring more voices to the table in studying the impact of these LLMs. Definitions of risk, harm, bias, and toxicity, etc., should be articulated by the collective research community as a whole, which is only possible when models are available for study.

We are releasing all of our models between

Many recent language models, such as ChatGPT, are only accessible via APIs, and the full set of learned parameters is not disclosed.

OPT (Open Pre-trained Transformer Language Models):

- Decoder-based models available in various sizes, ranging from 125M to 175B parameters.
- Models with 125M to 66B parameters are fully open to the public, while the 175B parameter model is restricted to research purposes only.

Past research and techniques

4. OPT (2022, Meta AI)

Category	GPT-3	OPT-175B
Gender	**62.6**	65.7
Religion	73.3	**68.6**
Race/Color	**64.7**	68.6
Sexual orientation	**76.2**	78.6
Age	**64.4**	67.8
Nationality	**61.6**	62.9
Disability	**76.7**	**76.7**
Physical appearance	**74.6**	76.2
Socioeconomic status	**73.8**	76.2
Overall	**67.2**	69.5

Table 4: **CrowS-Pairs evaluation.** Lower is better for all categories, indicating more fairness. The OPT-175B model performs worse than Davinci in most categories.

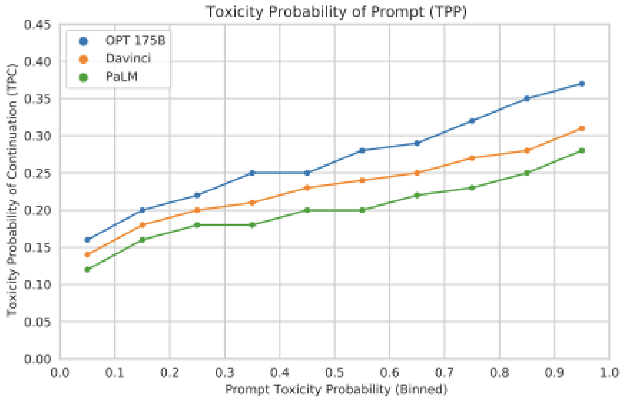

Figure 5: **RealToxicityPompts**. OPT-175B is more likely to generate toxic responses than either Davinci or PaLM. Consistent with prior work, toxicity rates increase as prompt toxicity increases.

Bias Exists :
- Hate speech and stereotypes are detectable.
- The model may also generate harmful content.

Limitations :
- Performs poorly on declarative instructions or direct questions.
- May generate factually incorrect statements, biased content, or harmful sentences.

Significance :
- An LLM that fully discloses its model parameters.
- Demonstrates performance comparable to GPT-3 while producing only 1/7 of its carbon footprint.
- Shares insights into various challenges encountered during the training process.

Past research and techniques

5. PaLM (2022, Google Research)

Model	# of Parameters (in billions)	Accelerator chips	Model FLOPS utilization
GPT-3	175B	V100	21.3%
Gopher	280B	4096 TPU v3	32.5%
Megatron-Turing NLG	530B	2240 A100	30.2%
PaLM	540B	6144 TPU v4	46.2%

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Task	0-shot		1-shot		Few-shot	
	Prior SOTA	PaLM 540B	Prior SOTA	PaLM 540B	Prior SOTA	PaLM 540B
TriviaQA (EM)	71.3[a]	**76.9**	75.8[a]	**81.4**	75.8[a] (1)	**81.4** (1)
Natural Questions (EM)	**24.7**[a]	21.2	26.3[a]	**29.3**	32.5[a] (1)	**39.6** (64)
Web Questions (EM)	**19.0**[a]	10.6	**25.3**[b]	22.6	41.1[b] (64)	**43.5** (64)
Lambada (EM)	77.7[f]	**77.9**	80.9[a]	**81.8**	87.2[c] (15)	**89.7** (8)
HellaSwag	80.8[f]	**83.4**	80.2[c]	**83.6**	82.4[c] (20)	**83.8** (5)
StoryCloze	83.2[b]	**84.6**	84.7[b]	**86.1**	87.7[b] (70)	**89.0** (5)

we discuss the ethical considerations related to large language models and discuss potential mitigation strategies.

Does not utilize pipeline parallelism (e.g., layer-based parallelism).PaLM was trained using 6144 TPU chips, with 12-way model parallelism on 3072 chips and 256-way data parallelism.

Introduced Model FLOPs Utilization (MFU), which enables precise comparisons based on system efficiency, demonstrating that PaLM performs the most efficient computations.

By leveraging a large number of parameters, PaLM surpasses most existing few-shot-based SOTA models.

Past research and techniques

5. PaLM (2022, Google Research)

One downside of self-supervision has been the move towards uncurated data. Models may mirror misinformation, stereotypes and bias in the corpus (Sheng et al., 2019; Kurita et al., 2019; Dev et al., 2019; Blodgett et al., 2020; Sheng et al., 2021). This is undesirable for scientific tasks which value truth. Uncurated data also means more tokens with limited transfer value for the target use-case; wasting compute budget. For example, the PaLM corpus is 50% social media conversations, which may have limited transfer towards scientific tasks (Chowdhery et al., 2022). The properties of scientific text also differ from general text - e.g. scientific terms and mathematics - meaning a general corpus and tokenizer may be inefficient. We explore whether a normative approach to dataset selection can work with the large model paradigm in this work.

Limitations of PaLM (as identified in the Galactica study)

50% of PaLM's corpus consists of social media conversations, which may limit its transferability to scientific tasks.
The Galactica study explores whether a normative approach to dataset selection can work effectively alongside the large-scale model paradigm.

The Normative Approach :
a method for structuring or selecting datasets by establishing normative criteria to manage data systematically and goal-orientedly. Instead of merely collecting large volumes of data, this approach focuses on constructing datasets selectively based on their quality, context, and relevance

Galactica surpasses numerous existing models in performance.

Dataset	Domain	GAL	OPT	BLOOM	GPT-3	Gopher	Chinchilla
Abstract Algebra	*out-of-domain*	**33.3%**	21.0%	25.0%	-	25.0%	31.0%
ARC Challenge	*in-domain*	**67.9%**	31.1%	32.9%	51.4%	-	-
ARC Easy	*in-domain*	**83.8%**	37.4%	40.7%	68.8%	-	-
Astronomy	*out-of-domain*	65.1%	23.0%	25.7%	-	65.8%	**73.0%**
BioASQ	*in-domain*	**94.3%**	81.4%	91.4%	-	-	-
Biology (College)	*out-of-domain*	68.8%	30.6%	28.5%	-	70.8%	**79.9%**
Biology (High-School)	*out-of-domain*	69.4%	27.7%	29.4%	-	71.3%	**80.3%**
Chemistry (College)	*out-of-domain*	46.0%	30.0%	19.0%	-	45.0%	**51.0%**
Chemistry (High-School)	*out-of-domain*	47.8%	21.7%	23.2%	-	47.8%	**58.1%**
Comp. Science (College)	*out-of-domain*	49.0%	17.0%	6.0%	-	49.0%	**51.0%**
Comp. Science (High-School)	*out-of-domain*	**70.0%**	30.0%	25.0%	-	54.0%	58.0%
Econometrics	*out-of-domain*	42.1%	21.0%	23.7%	-	**43.0%**	38.6%
Electrical Engineering	*out-of-domain*	**62.8%**	36.6%	32.4%	-	60.0%	62.1%
Elementary Mathematics	*out-of-domain*	38.1%	25.7%	27.6%	-	33.6%	**41.5%**
Formal Logic	*out-of-domain*	32.5%	29.4%	26.2%	-	**35.7%**	33.3%
Machine Learning	*out-of-domain*	38.4%	28.6%	25.0%	-	41.1%	41.1%
Mathematics (College)	*out-of-domain*	**43.0%**	33.0%	25.0%	-	37.0%	32.0%
Mathematics (High-School)	*out-of-domain*	**32.6%**	24.4%	27.0%	-	23.7%	31.9%
Medical Genetics	*out-of-domain*	**70.0%**	35.0%	36.0%	-	69.0%	69.0%
Physics (College)	*out-of-domain*	42.2%	21.6%	18.6%	-	34.3%	**46.1%**
Physics (High-School)	*out-of-domain*	33.8%	29.8%	25.2%	-	33.8%	**36.4%**
MedQA-USMLE	*out-of-domain*	44.4%	22.8%	23.3%	-	-	-
MedMCQA Dev	*in-domain*	**52.9%**	29.6%	32.5%	-	-	-
PubMedQA	*in-domain*	**77.6%**	70.2%	73.6%	-	-	-
Statistics (High-School)	*out-of-domain*	41.2%	43.5%	19.4%	-	50.0%	**58.8%**

Table 10: Question Answering Results. Galactica is evaluated without few-shot examples. Other LLMs are evaluated 5-shot, except for 0-shot results for GPT-3 on ARC results and OPT and BLOOM on PubMedQA and BioASQ. For abstract algebra and medical genetics, we obtained best results with 30B, so we report these scores; the 120B scores for these were 27.0% and 68.0% respectively. Rest of results are for 120B.

- Galactica outperforms larger models in most tasks and demonstrates superior performance in mathematical and graduate-level tasks.
- However, its corpus is biased toward graduate-level scientific knowledge, which sometimes leads to underperformance.
- In the remaining tasks, it achieves state-of-the-art results compared to fine-tuned models.

Necessity of Galactica

- Overwhelming Volume of Scientific Knowledge
- Limitations of Search Engines and Current Models
- Efficient Knowledge Representation and Reasoning
- Domain-Specific Performance
- Unification of Modalities
- Accelerating Research and Discovery

Galactica addresses the challenges of modern scientific research by providing an advanced tool that combines domain-specific knowledge, reasoning capabilities, and efficient data synthesis, paving the way for more streamlined and innovative scientific workflows.